

FORTUNA

Research & Analysis Platform

FORMAL SPECIFICATION & DEVELOPMENT DOCUMENT · v0.4

Project:	Fortuna — R&A (Research & Analysis)
Version:	v0.4 — Direction Funnel Re-Index (Track A) + Risk Management
Owner:	Ash
Stack:	FastAPI · Supabase · Claude API · TradingView · VectorBT
Broker:	Dhan HQ v2 - Trading APIs (free) + Data API (Phase 2+)
Date:	July 12, 2026

Changelog

Versi on	Date	Key Changes
v0.1	Jul 2026	Initial skeleton — 10 modules, single-tool concept, local machine
v0.2	Jul 2026	Three-tool architecture, model decisions, index-first strategy, Render hosting
v0.3	Jul 2026	Direction Engine, Foundation List v0.1 (26 items), Risk Management, Dhan MCP mapped, full stack finalised
v0.4	Jul 12, 2026	Track A re-index: Foundation List restructured into 5-stage Direction Funnel (MAC/NAT/FLO/PRE/ENG). 5 new items discovered (PRE-4 CPR, ENG-1 Regime, ENG-5 Supertrend, ENG-9 ATR, ENG-10 Bollinger). 19 of 22 funnel items researched with formal recommendations. Coding pipeline + two-book branch filters documented.

1. Project Overview

Fortuna is a personal trading research and analysis platform for Indian equity and index options markets. It aggregates multi-layer market data, performs technical and fundamental analysis, applies the trader's own recorded intelligence via AI, and surfaces a directional view every morning — accessible from laptop and mobile via browser.

Foundation Principle: Every architectural decision must support scale. We build the basement of a skyscraper, not a bungalow.

1.1 Two Active Trading Modes

- Nifty50 intraday options — 5min chart, direction decided by morning analysis, all trades placed manually by Ash
- Swing equity (Fortuna-Cash) — multi-day, filtered by the same Direction Engine foundation list

1.2 Development Protocol

- No code written without prior formal approval
- Every module: Proposal → Approval → Code → Review → Next
- Bugs spotted outside current scope: reported first, fixed only after approval
- Nothing goes into code without explicit approval in the current task

TRADE EXECUTION RULE: Ash places all trades manually on Dhan. Claude does not place trades at any stage of development unless explicitly approved in a future phase.

2. Three-Tool Architecture

Fortuna is three distinct tools under one platform, sharing the same infrastructure, database, AI layer, Direction Engine, and Dhan API connection.

FORTUNA-INDEX

Index Research & Options Analysis

Nifty50, BankNifty and other indexes — direction engine, options chain, all indicators, chart with drawing tools. FIRST TO BUILD.

- Full Direction Engine output every morning — Bull / Bear / Sideways with confidence score
- All 26 foundation list items computed with individual signals
- Options chain analysis — PCR, max pain, OI buildup, IV, full Greeks
- TradingView Lightweight Chart with pre-marked S&R; levels — Ash draws additional levels manually
- Risk management panel — daily SL and max loss shown before any trade is placed

FORTUNA-CASH

Individual Stock Swing Trading

Nifty500 equity screening, fundamentals, scoring, portfolio tracking. SECOND TO BUILD.

- Same Direction Engine as first filter — only trade in direction of index trend
- Full fundamental + technical scoring across Nifty 500 universe
- Sector rotation and FII/DII flow context per stock
- Trade journal, P&L; tracking, and rolling performance analytics

FORTUNA-OPTIONS

Individual Stock Options

To be specced after Index and Cash tools are stable in live use. Liquidity concerns noted.

3. Direction Engine

The Direction Engine is now organised as a daily direction funnel (Track A v0.7) — broadest macro scan at the top, narrowing through national news, institutional flows, and pre-market gates, into an 11-item technical engine, producing a single index-direction verdict. That verdict then branches into the two trading books: Index Options and Swing/Stock.

3.1 Output Format — Dual View

SYSTEM SCORE

Each funnel item is computed and scored individually. All signals cascade top-down through the funnel into a weighted directional score — e.g. BEARISH 7/10. Ash can agree or override any individual item before the final view is formed.

ASH'S OWN VIEW

Ash enters a personal directional view per item and an overall bias. Stored in Supabase alongside the system score. Claude reads both views and synthesises them into the morning briefing commentary.

Both views are displayed side by side. Claude synthesises them into a single directional commentary. The final trading decision is always Ash's.

3.2 The Direction Funnel — Five Stages

Every item now carries a permanent funnel ID (e.g. MAC-1) with the original research ID kept as an alias (e.g. was B4) so references stay consistent across every document.

Stage	Prefix	Items	Status
1 · Macro scan	MAC	MAC-1..3 (B4/B6/B12)	DONE
2 · National news	NAT	NAT-1 (B5)	DONE
3 · Institutional flows	FLO	FLO-1..2 (C1/C2)	DONE
4 · Pre-market gates	PRE	PRE-1..5 (VIX/GIFT/gap/CPR/season)	3 of 5
5 · Direction engine	ENG	ENG-1..11 (technical block)	DONE

Output: Index Direction (Bull / Bear / Sideways + confidence) → branches into Branch A (Index Options: PCR, max pain, OI, expiry) and Branch B (Swing/Stock: sector, screen, relative strength, fundamentals).

3.3 Stage 1 — Macro Scan

Global risk tone — the widest layer. Fed, US/Asia markets, DXY/US10Y/crude/gold.

MAC-1	External Markets	Y* — Include w/ caveats	Global macro news (was B4) Event gate / regime overlay. Neutral ~90% of days, dominant on rare event days. Edge is at the open, decays within 30-45 min. Fed/geo events are directional via FII flow; crude shocks move volatility more than direction.
MAC-2	External Markets	Y — Include	US / Asian market correlation (was B6) The most codeable macro item. Nifty-S&P; 30-day correlation reached 0.68. Edge is at the open: GIFT implied gap + US close + Asian confirmation, gated by VIX (12-16 = trade; >18 = fade).
MAC-3	External Markets	Y* — Include w/ caveats	Macro levels: DXY/US10Y/crude/gold (was B12) Mirror image of intraday items — strong at weekly/monthly, near-zero intraday. DXY and US10Y are genuine positional drivers (threshold-based). Gold is not a predictor ($R^2 < 0.3\%$) — drop as signal, keep as risk-off confirmation only.

3.4 Stage 2 — National News

Domestic event flags — RBI, Union Budget, elections, regulatory policy.

NAT-1	External Markets	Y* — Include w/ caveats	National news: RBI/Budget/elections (was B5) Scheduled-event gate. Output for options buyer is trade/no-trade + IV-crush warning, not continuous weight. Elections (surprise only) can move Nifty -5.93% in a day. Budget: no reliable direction, but VIX crush kills long options on all event days.
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3.5 Stage 3 — Institutional Flows

Realised flow + leveraged intent — the FII/DII layer.

FLO-1	Institutional	Y* — Include w/ caveats	FII/DII daily cash flow (was C1) Cash data publishes after close (6-7 PM) — structurally next-day/swing, never intraday. Single-day $R^2 \sim 16.5\%$; 3-5 day rolling trend is the reliable read. Read the NET (FII+DII), never FII alone.
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FLO-2	Institutional	Y — Include	FII index derivatives positioning (was C2) EOD data (~7 PM) — next-day/multi-week signal. Best read is divergence: retail buying calls at highs while FIIs liquidate longs = trap-reversal signal. Always cross-check FLO-1 to separate a directional short from a hedge.
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3.6 Stage 4 — Pre-Market Gates

The 9:00 AM scan — produces a trade / no-trade decision and day-type classification before the open.

PRE-1	Patterns	PENDING RESEARCH	India VIX regime (was G5) Master volatility gate: <13 trending/reliable, >18 fade, >20 avoid directional buying. Sets trade/no-trade and premium context for both books. Was mis-filed under Patterns — functionally a pre-market gate.
PRE-2	Patterns	PENDING RESEARCH	GIFT Nifty pre-market (was G7) The 6:30 AM opening-direction predictor consumed by MAC-2 and PRE-3. ~75-85% accuracy on opening direction, degrades on event days. Was mis-filed under Patterns.
PRE-3	Price Structure	Y — Include (strong)	Gap up / gap down behaviour (was E9) Gap direction alone is near a coin flip (29.4% same-day fill). Edge is gap + trend-alignment + ATR-normalised size + VIX + no-news. Four-type taxonomy: Common (noise), Runaway (continues 55-65%), Breakaway (trends, rarely fills), Exhaustion (reverses ~78%).
PRE-4	Technical	Y — Include (strong)	CPR — Central Pivot Range (was D8-11) NEW ITEM. Three levels (BC/PP/TC) from prior day H/L/C, known before open. CPR width forecasts day-type: narrow (<35pts) = trend day, wide (>60pts) = range day. Front-runs the coincident regime gate ENG-1.
PRE-5	Backtesting	PENDING RESEARCH	Day-of-week / seasonality (was A7) Monday-gap bullishness (58.3% close higher), expiry-week bias, month-end FII rollover. Was mis-filed under Backtesting — functionally a pre-market calendar tilt read.

PRE-1, PRE-2, and PRE-5 are the next research targets in Track A / Track C. If already researched in the Track C chat, pull those conclusions in rather than duplicating.

3.7 Stage 5 — Direction Engine (11 Items)

Regime to trend to timing to S/R — the technical cascade. This is also the Module I-8 dashboard order.

ENG-1	Technical	Y — Include (strong)	Regime filter: ADX + Choppiness (was D8-7) NEW ITEM. The master gate every other engine item assumes. 0 directional weight — it decides whether the cascade should be trusted at all. $CHOP < 38.2 + ADX > 25 =$ trend (full weight); $CHOP > 61.8 =$ chop (stand aside).
ENG-2	Technical	Y — Include (strong)	EMAs across timeframes (was D8-6) The structural backbone. EMA stack + slope per TF = that TF's trend definition. 20/50/200 = dynamic S/R where pullback entries bounce. Crossovers are lagging confirmation only. Test 13-21-34 (see A2).
ENG-3	Technical	Y — Include (strong)	RSI — Cardwell ranges (was D8-2) Stronger and more leading than MACD via Cardwell ranges (bull 40-80, bear 20-60), not naive 70/30. Range-shift leads price. Ceiling rejection = exit (not buy put) — matches Ash's existing framework exactly.
ENG-4	Technical	Y* — Include w/ caveats	MACD crossover (was D8-1) A lagging, price-derived tool. Demoted from primary to confirmation. Value comes from higher-TF MACD as trend filter — raw 5-min crossover is the weakest, most whipsaw-prone form.
ENG-5	Technical	Y* — Include w/ caveats	Supertrend (was D8-8) NEW ITEM. ATR-based trend line, India's most-used intraday indicator. Clean trend filter (only longs green, shorts red), not an every-flip trigger (raw flips = 45-60% win). Doubles as volatility-adaptive trailing SL.
ENG-6	Technical	Y — Include (trigger only)	Stochastic RSI (was D8-3) Fastest but noisiest oscillator — pure entry-timing trigger, not direction. RSI decides which way; StochRSI decides the exact tick. Gated hardest of all items.

ENG-7	Technical	Y — Include (strong)	VWAP + SL decisions (was D8-4) The intraday complement to EMA. Above/below VWAP = call/put bias. Hub of the four-SL framework (VWAP SL, Hunt Zone, Structure SL, ATR SL). Formalises Ash's G8 SL-hunt observation directly.
ENG-8	Technical	Y — Include (2 roles)	Volume & liquidity (was D8-5) Two signals: (1) volume-as-confirmation = conviction multiplier, (2) liquidity-as-gate = binary tradeability filter. Stock options must clear a 4-metric liquidity screen; index auto-passes.
ENG-9	Technical	Y — Include (infrastructure)	ATR — volatility backbone (was D8-10) NEW ITEM. Volatility backbone, not a direction signal. Feeds Supertrend bands, hunt zone, SL sizing, position sizing. 0 directional weight, essential plumbing. Must be tracked per-instrument.
ENG-10	Technical	Y — Include (squeeze only)	Bollinger Bands + squeeze (was D8-9) NEW ITEM. Keep the squeeze (bands compress = big move loading), drop band-touch mean-reversion (fails in trends). Squeeze tells you WHEN, histogram tells you WHICH WAY. Ideal for buying cheap premium before expansion.
ENG-11	Technical	Y — Include (strong)	Support & Resistance + OI walls (was E10) The "where" map. Not a direction signal — defines where entries trigger, moves stall, pullbacks bottom. Entire edge is confluence (3+ level types stacking = A+ zone). Includes Call/Put walls, max pain, change-in-OI.

3.8 Direction Sequence — Coding Pipeline

A gate-first, top-down pipeline where each stage can veto the next. Each higher timeframe (Monthly → Weekly → Daily → 4H → 1H → 15min → 5min) gates the one below.

#	Stage	Question	Items	Gate / Action
1	Regime gate	Trend or chop?	ENG-1 + PRE-4	Chop / wide CPR → STOP. Runs first.
2	Higher-TF bias	Which way?	ENG-2 + ENG-3	Sets dominant direction. Conflicting → wait.
3	Direction confirm	Momentum agrees?	ENG-4 + ENG-5	Must agree with step 2, else no trade.

4	Location (where)	Pullback bottom?	ENG-11+ENG-7+2	Confluence zone (3+ types) = entry area.
5	Timing (when)	Turn happening?	ENG-6	StochRSI out of OS/OB at the zone = trigger.
6	Conviction	Move real?	ENG-8+9+10	Weak volume / no expansion → skip or size down.
7	Risk (stop)	Where's the exit?	ENG-7+5+9	SL from VWAP / Supertrend / structure ± ATR.

THE RULE: regime → bias → direction → location → timing → conviction → stop. Pre-open, PRE-3 (gap) sets the first-hour bias. A bull cascade buys CALLs on the with-trend trigger; the counter-trend signal is an exit/wait, never a PUT.

3.9 The Two Books — Branch Filters

After index direction is set, each trading book applies its own further filters. These are the next research targets.

Branch	Prefi x	Filters (in order)	Output
A · Index Options	OPT	Options intelligence (PCR, max pain, OI, IV, Greeks) → expiry behaviour → 5-min execution + VWAP SL-hunt	Strike + CE/PE
B · Swing / Stock	STK	Index direction → sector rotation → NSE screen → relative strength → fundamentals → stock cascade → liquidity gate	Stock swing / option

Foundation List — 26 core items re-indexed into a 5-stage funnel plus 5 new items discovered during research (PRE-4/ENG-1/5/9/10). 19 of 22 funnel items DONE. Pending: PRE-1 (VIX), PRE-2 (GIFT), PRE-5 (seasonality) — then Branch A/B filters.

4. Risk Management Framework

Risk management parameters are calculated and displayed by Fortuna before any trade is placed each session. All figures are informational — Ash makes the final decision on every trade. No automatic exits, no forced stop-outs.

TRADE EXECUTION RULE: Ash places all trades manually on Dhan. Claude does not place trades at any stage of development unless explicitly approved in a future phase.

4.1 Daily Trading Rules

Rule	Value	Rationale
Max trades per day	2 trades	Discipline enforced. STOP banner shown after 2 trades regardless of outcome.
Hard limit	3 trades	Absolute maximum. Never exceeded under any circumstance.
Daily max loss	Ash-set %	Stored in Supabase as % of capital. Fortuna displays warning at threshold.
Per-trade risk	Auto-calc	Daily limit divided by max trades. Shown as maximum lot size guidance before each trade.

4.2 The SL Problem — Ash's Observation

Book SL (ATR-based or fixed %) does not work well for Nifty intraday options. Price regularly falls below VWAP to hit all retail stop losses, then reverses and continues the original direction. A rigid SL gets triggered at exactly the wrong moment — the hunt bottom. This is documented market maker behaviour on Nifty.

4.3 SL Reference Levels — All Displayed on Chart

SL Type	Calculation	When to Use
ATR SL	Entry minus $1.5 \times \text{ATR}(14)$ on 5min chart	Starting reference only. Use as initial position size guide, not a hard stop.
VWAP SL	Below current VWAP lower band	2 consecutive candle closes below VWAP = exit watch. One close = observe only.
Structure SL	Below last swing low on 5min chart	Hard stop — if this level breaks, the intraday trend structure has changed.
Hunt Zone	VWAP minus $0.5 \times \text{ATR}$ on 5min	Stay in trade if price is only within hunt zone. Exit only if price goes deeper.

Time SL	Exit after 30 min with no move	Prevents premium decay trap in sideways sessions. Time is the enemy of options buyers.
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4.4 VWAP SL Decision Logic — Formalised

- Price closes below VWAP on 5min → watch mode. Do not exit immediately.
- Price closes 2 consecutive candles below VWAP → SL hunt in progress. Watch for reversal.
- Price reclaims VWAP (closes above) → stay in trade. Hunt is over. Direction continues.
- Price reclaims VWAP then takes resistance at VWAP from above → VWAP has flipped to resistance. Exit or reverse.
- Price goes deeper than VWAP minus $0.5 \times \text{ATR}$ → hunt unusually deep. Move to structure SL.

Fortuna shows all four SL levels on the chart pre-marked as coloured lines. Ash watches the candles and decides which level applies. Fortuna never automatically triggers an exit.

5. AI Model Decisions

Purpose	Model	Reason
Writing Fortuna code	Claude Opus 4.6	Complex architecture, multi-file reasoning, highest code quality required
Daily Direction Engine AI runs	Claude Sonnet 4.6	Runs 2-4× daily — balance of reasoning depth and cost (~Rs.1-2 per full run)
News tracking and investor sentiment	Claude Sonnet 4.6	Frequent structured calls, cost sensitive, fast response required
FII prediction and macro research	Claude Opus 4.6	Deep multi-factor reasoning, run occasionally not daily
Backtesting logic and pattern analysis	Claude Opus 4.6	Complex historical pattern reasoning, depth over cost
Planning and architecture discussions	Claude Sonnet 4.6	Fast, accurate, cost effective for ongoing discussion

6. Technology Stack

Layer	Technology	Purpose	Cost
Backend framework	FastAPI (Python)	Async API, native WebSocket, concurrent laptop + mobile requests	Free
Database	Supabase	Foundation list, trader notes, trade journal, scores, 24h cache	Free tier
Technical indicators	pandas-ta-classic + TA-Lib	284 indicators + 62 candlestick patterns, Numba-accelerated	Free
Backtesting	VectorBT (open source)	NSE-native, Indian fee models built in, 1000× faster than alternatives	Free
Charting	TradingView Lightweight Charts	Candlestick charts, drawing tools, indicator overlays — mobile-ready	Free
AI layer	Claude API (Sonnet + Opus)	Direction Engine synthesis, news analysis, trader notes application	~Rs.1-2/r un
Broker data	Dhan Historical + Live APIs	Clean NSE data from inception, options chain with Greeks, live feed	Rs.499/ mo Phase 2+
Broker trading	Dhan Trading APIs	Portfolio, positions, P&L;, funds, kill switch — all free tier	Free
Broker MCP	Dhan native MCP server	Natural language interface to Dhan account — read access confirmed	Free
Hosting	Render (free tier)	On-demand spin-up ~30 seconds cold start — 2-4 runs/day within free limit	Free
Access	Browser — laptop + mobile	Responsive design, same URL, no app install required	Free

Phase 1: yfinance + NSE scraping for price data (free). Phase 2 onwards (Fortuna-Index live): Dhan Data API Rs.499/month replaces yfinance entirely — cleaner NSE data, no split-adjustment issues.

7. Dhan Integration Map

7.1 What Claude Can Access via Dhan MCP (Read Access)

- Live holdings — every position, quantity, average cost price, T1 quantity
- Open positions — intraday and delivery, realised + unrealised P&L; live
- Available funds — free cash, margin used, collateral available
- Order book — all today's orders with status
- Trade history — all executed trades today
- Live LTP for any instrument — equity, F&O;, MCX
- Full option chain — all strikes, OI, Greeks (Delta, Theta, Gamma, Vega), IV, volume, bid/ask
- Historical daily OHLCV — from inception date, all NSE/BSE/MCX instruments
- Historical intraday OHLCV — 1, 5, 15, 25, 60 min candles — last 5 years
- Full 200-level market depth — demand and supply zone detection
- Expired options data — historical OI for all backtesting work

7.2 Trade Execution — Current Phase

Claude does not place trades. Ash places all trades manually on Dhan. This rule applies until a future phase is explicitly approved after paper trading validation is complete.

7.3 Cost Summary

API Category	Cost	Phase
Trading APIs (orders, portfolio, positions, funds, kill switch, P&L; exit)	Free	All phases
Dhan MCP server	Free	All phases
Historical + Live Data API	Rs.499+GST/month	Phase 2+
CNC equity brokerage	Free per trade	All phases
F&O; brokerage	Rs.20 per order	All phases

8. Module Inventory — Fortuna-Index

First tool to be built. Each sub-component requires an individual formal spec and Ash's approval before any code is written.

MODULE I-1 Data Engine — Index

All market data for indexes and their options. No other module fetches data directly.

I-1A	Index Universe Loader	Nifty50, BankNifty, FinNifty, MidcpNifty — symbol list and metadata, refreshed weekly
I-1B	Index OHLCV Puller	Historical and EOD price data across all timeframes — monthly down to 1min
I-1C	NSE Session Manager	Session headers and cookies for all NSE API calls — FIRST TO BUILD
I-1D	Options Chain Fetcher	Full chain — all strikes, OI, Greeks, IV, volume for any underlying
I-1E	Live Quote Fetcher	Current index level, day range, change — via Dhan Market Quote API
I-1F	FII Derivatives Fetcher	Participant-wise OI — FII net long/short in index futures and options
I-1G	Gift Nifty Fetcher	Pre-9:15 AM direction indicator from SGX/Gift Nifty feed
I-1H	External Market Fetcher	US markets, Asian markets, Gold, Crude oil, DXY, India VIX
I-1I	News Fetcher	RSS feeds — ET Markets, Moneycontrol, Reuters India — pre-market summary via Claude
Approval Status		PENDING

MODULE I-2 Direction Engine

Processes all 26 foundation list items, scores each signal, produces Bull/Bear/Sideways verdict.

I-2A	Foundation Item Scorer	Computes each A/B/C/D/E/F/G item signal — Bullish / Bearish / Neutral
I-2B	Timeframe Aggregator	Combines signals across monthly to 5min — higher timeframes carry more weight
I-2C	System Score Calculator	Weighted aggregate score on a 0–10 Bull to Bear scale
I-2D	Ash View Interface	Ash enters personal view per item and overall bias — stored in Supabase

I-2E	Synthesis Engine	Claude reads system score + Ash view + news + stored notes — writes directional commentary
I-2F	Regime Detector	Identifies current regime — trending / expiry week / volatile month / event day
Approval Status		PENDING

MODULE I-3 Technical Analysis Engine — Index

All indicators computed across all timeframes. pandas-ta-classic + TA-Lib acceleration.

I-3A	MACD — all timeframes	Crossover signals, histogram direction, state per timeframe
I-3B	RSI — all timeframes	Levels calibrated for Nifty index behaviour specifically
I-3C	Stochastic RSI — lower TFs	5min and 15min primary — crossover as entry confirmation signal
I-3D	VWAP — 1min, 5min, 1H	With upper/lower bands. Hunt zone = VWAP minus 0.5× ATR pre-calculated
I-3E	EMAs — 9, 20, 50, 200	All timeframes. EMA confluence zones pre-calculated and shown as S&R; on chart
I-3F	ATR — 14 period	Volatility measure, primary input for SL calculation across all timeframes
I-3G	Volume analysis	Today vs 20-day average. Volume on individual 5min candles tracked
I-3H	Supertrend	Trend direction with dynamic SL — daily and 1H timeframes primary
I-3I	India VIX	Current level, regime classification, change from previous session
Approval Status		PENDING

MODULE I-4 Options Intelligence Engine

Derives actionable intelligence from index options chain. Core of Fortuna-Index.

I-4A	PCR Calculator	Put-Call Ratio — OI based and volume based both tracked and displayed
I-4B	Max Pain Calculator	Strike where maximum options expire worthless — gravitational price level
I-4C	OI Buildup Analyser	Where OI is being added and shed — identifies real S&R; invisible on price chart

I-4D	IV Percentile	Current IV vs 1-year range — cheap or expensive options context
I-4E	Change in OI Tracker	Session-on-session OI change per strike — where smart money moved
I-4F	FII vs Retail Positioning	Participant-wise OI — are FIIs net long or short on index derivatives
I-4G	Intraday OI Change	Live OI change during the session — real-time institutional positioning signal
Approval Status		PENDING

MODULE I-5 Risk Management Engine

Daily risk parameters calculated and displayed before trading begins. Informational only — no forced exits.

I-5A	Capital Reader	Available funds fetched fresh from Dhan Funds API each morning
I-5B	Daily Loss Limit Display	Ash-set % of capital, stored in Supabase, shown prominently before first trade
I-5C	Per-Trade Risk Calculator	Daily limit divided by max trades — shown as maximum lot size guidance
I-5D	Multi-SL Calculator	ATR SL, VWAP SL, Structure SL, Hunt Zone — all four levels shown as chart lines
I-5E	Trade Counter	Tracks trades taken today — STOP banner at 2 trades, hard limit at 3
I-5F	Running P&L; Display	Live P&L; from Dhan Positions API, updated throughout the trading session
Approval Status		PENDING

MODULE I-6 Chart Engine

Interactive candlestick chart with drawing tools. TradingView Lightweight Charts — mobile-ready.

I-6A	Candlestick Chart	OHLCV candlestick chart — multiple timeframes switchable by Ash
I-6B	Indicator Overlay	EMA lines, VWAP, Supertrend, Bollinger Bands plotted directly on chart
I-6C	Drawing Tools	Trendlines, horizontal S&R; lines, Fibonacci retracement — Ash draws manually
I-6D	Pre-marked Levels	Previous day/week high-low, max pain, high OI strikes — auto-drawn by Fortuna

I-6E	SL Levels Overlay	ATR SL, VWAP SL, Structure SL, Hunt Zone — shown as distinct coloured lines
I-6F	Pattern Annotation	Detected candlestick patterns highlighted automatically on the chart
Approval Status		PENDING

MODULE I-7 AI Analysis Layer — Index

Claude reads all data + Ash's stored notes + news + both views. Produces directional commentary.

I-7A	Trader Notes Storage	Ash records personal price action observations per market condition — Supabase
I-7B	Context Builder	Packages all module outputs + trader notes into structured Claude API prompt
I-7C	Direction Commentary	Claude synthesises system score + Ash view + news into morning briefing text
I-7D	SL Hunt Pattern Detector	Claude flags when current price action matches the SL hunt pattern from notes
I-7E	Post-trade Learning	After trade closes, Claude asks what happened — builds personal pattern library over time
Approval Status		PENDING

MODULE I-8 Report Generator — Index

Single HTML dashboard assembled from all modules. Browser-based, responsive for laptop and mobile.

I-8A	Morning Briefing Panel	Direction score, each item signal colour-coded, Ash view input, Claude commentary
I-8B	Risk Panel	Capital available, daily limit, per-trade limit, trade counter, running P&L;
I-8C	Options Intelligence Panel	PCR, max pain level, OI heatmap by strike, IV percentile, FII positioning
I-8D	Chart Panel	TradingView chart with all pre-marked levels, SL lines, and drawing tools
I-8E	External Markets Panel	US market close, Asian markets open, Gift Nifty, Gold, Crude oil, DXY
I-8F	Run Metadata	Timestamp of last run, data freshness indicators, any fetch failures clearly flagged
Approval Status		PENDING

9. Research Notes

Open research threads. Findings from each note feed directly into Foundation List item weights or new module additions. Nothing is coded from a research note without a separate formal spec and Ash's approval.

RN-001	Stocks that move 20%+ in a single session	<i>OPEN — Under Investigation</i>
RN-002	Candlestick patterns, price patterns, trader price action logic	<i>OPEN — Partially Mapped</i>
RN-003	Historical analysis and backtesting engine	<i>OPEN — Phase 2</i>
RN-004	FII prediction — anticipating inflows before they hit price	<i>OPEN — To Be Researched</i>
RN-005	Forex and Asian market crash prediction	<i>OPEN — To Be Researched</i>
RN-006	Retail exit patterns during market stress	<i>OPEN — To Be Researched</i>
RN-007	SIP, ETF, IPO research buttons — three one-click tools	<i>OPEN — Future Module</i>
RN-008	Foundation List — deep research pass on all 26 items (Track A)	<i>OPEN — PRIORITY</i>

10. Cost Structure

Component	Tool	Phase 1	Phase 2+
EOD Price Data	yfinance	Free	Replaced by Dhan
Intraday + Historical Data	Dhan Data API	Not needed	Rs.499+GST/mo
Option Chain	NSE API / Dhan	Free	Free
FII/DII Data	NSE scraping	Free	Free
Trading + Portfolio APIs	Dhan (free tier)	Free	Free
Dhan MCP server	Dhan native MCP	Free	Free
Database	Supabase (free tier)	Free	Free
Hosting	Render (free tier)	Free	Free
AI — Daily runs	Claude Sonnet 4.6	~Rs.1-2/run	~Rs.1-2/run
AI — Deep research	Claude Opus 4.6	On demand	~Rs.5-10/run
CNC Equity Brokerage	Dhan	Free/trade	Free/trade
F&O; Brokerage	Dhan	Rs.20/order	Rs.20/order

11. Phase Roadmap

Phase 1A	Fortuna-Index	CURRENT	Direction Engine + chart + options intelligence + risk management. All modules I-1 through I-8. Manual trading by Ash throughout.
Phase 1B	Fortuna-Cash	NEXT	Equity swing screening, fundamentals, scoring. Same Direction Engine applied as the first filter before any stock qualifies.
Phase 2	Deep Research Pass	PLANNED	RN-008: research every Foundation List item. Assign weights validated by backtesting. One item per session on Track A.
Phase 3	Paper Trading Engine	PLANNED	Simulated trades, paper P&L; tracking, full strategy validation before any real automation is introduced.
Phase 4	Live Auto-Execution	FUTURE	Only after Phase 3 proves positive P&L; across trending, rotating, and risk-off market conditions.
Phase 5	Commodity Extension	FUTURE	MCX instruments added to the same pipeline. Same scoring engine, different universe — Gold, Crude, Base metals.